## ANSWER KEY

### English Test

| | | | |
|---|---|---|---|
| 1. A | 21. D | 41. A | 61. A |
| 2. F | 22. F | 42. H | 62. G |
| 3. C | 23. B | 43. D | 63. D |
| 4. H | 24. H | 44. F | 64. H |
| 5. A | 25. D | 45. D | 65. B |
| 6. G | 26. H | 46. J | 66. H |
| 7. C | 27. A | 47. A | 67. C |
| 8. F | 28. J | 48. G | 68. J |
| 9. D | 29. D | 49. D | 69. A |
| 10. G | 30. H | 50. H | 70. G |
| 11. D | 31. B | 51. B | 71. C |
| 12. G | 32. F | 52. J | 72. J |
| 13. B | 33. A | 53. C | 73. A |
| 14. F | 34. J | 54. F | 74. J |
| 15. C | 35. A | 55. C | 75. A |
| 16. G | 36. J | 56. F | |
| 17. A | 37. B | 57. B | |
| 18. F | 38. H | 58. G | |
| 19. B | 39. A | 59. C | |
| 20. F | 40. G | 60. J | |

## Mathematics Test

| | | |
|---|---|---|
| 1. C | 21. B | 41. B |
| 2. J | 22. F | 42. K |
| 3. B | 23. E | 43. E |
| 4. G | 24. H | 44. K |
| 5. B | 25. B | 45. A |
| 6. G | 26. K | 46. G |
| 7. E | 27. D | 47. A |
| 8. J | 28. H | 48. F |
| 9. E | 29. B | 49. B |
| 10. F | 30. K | 50. G |
| 11. B | 31. E | 51. E |
| 12. K | 32. F | 52. K |
| 13. B | 33. D | 53. B |
| 14. F | 34. G | 54. J |
| 15. C | 35. B | 55. B |
| 16. H | 36. F | 56. H |
| 17. B | 37. C | 57. E |
| 18. G | 38. G | 58. K |
| 19. D | 39. E | 59. E |
| 20. H | 40. G | 60. F |

## Reading Test

| | |
|---|---|
| 1. B | 21. A |
| 2. J | 22. F |
| 3. A | 23. D |
| 4. H | 24. G |
| 5. A | 25. D |
| 6. F | 26. H |
| 7. C | 27. B |
| 8. G | 28. F |
| 9. C | 29. C |
| 10. J | 30. J |
| 11. A | 31. A |
| 12. H | 32. G |
| 13. D | 33. C |
| 14. F | 34. F |
| 15. C | 35. D |
| 16. G | 36. H |
| 17. B | 37. A |
| 18. H | 38. G |
| 19. D | 39. D |
| 20. F | 40. J |

## Science Reasoning Test

| | |
|---|---|
| 1. D | 21. C |
| 2. G | 22. J |
| 3. B | 23. B |
| 4. J | 24. F |
| 5. B | 25. B |
| 6. F | 26. J |
| 7. A | 27. A |
| 8. F | 28. J |
| 9. C | 29. A |
| 10. G | 30. H |
| 11. D | 31. D |
| 12. H | 32. G |
| 13. B | 33. A |
| 14. H | 34. H |
| 15. C | 35. C |
| 16. F | 36. G |
| 17. B | 37. A |
| 18. G | 38. H |
| 19. C | 39. B |
| 20. J | 40. H |

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## SCORING GUIDE

Your final reported score is your COMPOSITE SCORE. Your COMPOSITE SCORE is the average of all of your SCALE SCORES.

Your SCALE SCORES for the four multiple-choice sections are derived from the Scoring Table on the next page. Use your RAW SCORE, or the number of questions that you answered correctly for each section, to determine your SCALE SCORE. If you got a RAW SCORE of 60 on the English test, for example, you correctly answered 60 out of 75 questions.

**Step 1** Determine your RAW SCORE for each of the four multiple-choice sections:

**English** _____

**Mathematics** _____

**Reading** _____

**Science Reasoning** _____

The following Raw Score Table shows the total possible points for each section.

| RAW SCORE TABLE | |
|---|---|
| **KNOWLEDGE AND SKILL AREAS** | **RAW SCORES** |
| ENGLISH | 75 |
| MATHEMATICS | 60 |
| READING | 40 |
| SCIENCE REASONING | 40 |
| WRITING | 12 |

## *Multiple-Choice Scoring Worksheet*

**Step 2** Determine your SCALE SCORE for each of the four multiple-choice sections using the following Scoring Worksheet. Each SCALE SCORE should be rounded to the nearest number according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32. If you answered 61 questions correctly on the English section, for example, your SCALE SCORE would be 29.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **English** | _____ RAW SCORE | × **36** = | _____ | ÷ **75** = | _____ | **− 2** | (*correction factor) |
| | | | | | _____ SCALE SCORE | | |
| **Mathematics** | _____ RAW SCORE | × **36** = | _____ | ÷ **60** = | _____ | **+ 1** | (*correction factor) |
| | | | | | _____ SCALE SCORE | | |
| **Reading** | _____ RAW SCORE | × **36** = | _____ | ÷ **40** = | _____ | **+ 2** | (*correction factor) |
| | | | | | _____ SCALE SCORE | | |
| **Science Reasoning** | _____ RAW SCORE | × **36** = | _____ | ÷ **40** = | _____ | **+ 1.5** | (*correction factor) |
| | | | | | _____ SCALE SCORE | | |

*The correction factor is an approximation based on the average from several recent ACT tests. It is most valid for scores in the middle 50% (approximately 16–24 scale composite score) of the scoring range.

The scores are all approximate. Actual ACT scoring scales vary from one administration to the next based upon several factors.

If you take the optional Writing Test, you will need to combine your English and Writing scores to obtain your final COMPOSITE SCORE. Once you have determined a score for your essay out of 12 possible points, you will need to determine your ENGLISH/WRITING SCALE SCORE, using both your ENGLISH SCALE SCORE and your WRITING TEST SCORE. The combination of the two scores will give you an ENGLISH/WRITING SCALE SCORE, from 1 to 36, that will be used to determine your COMPOSITE SCORE mentioned earlier.

Using the English/Writing Scoring Table, find your ENGLISH SCALE SCORE on the left or right hand side of the table and your WRITING TEST SCORE on the top of the table. Follow your ENGLISH SCALE SCORE over and your WRITING TEST SCORE down until the two columns meet at a number. This number is your ENGLISH/WRITING SCALE SCORE and will be used to determine your COMPOSITE SCORE.

**Step 3** Determine your ENGLISH/WRITING SCALE SCORE using the English/Writing Scoring Table on the following page:

**English** _____

**Writing** _____

**English/Writing** _____

## ENGLISH/WRITING SCORING TABLE

| ENGLISH SCALE SCORE | WRITING TEST SCORE | | | | | | | | | | | ENGLISH SCALE SCORE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | |
| **36** | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 32 | 36 | **36** |
| **35** | 26 | 27 | 28 | 29 | 30 | 31 | 31 | 32 | 33 | 34 | 35 | **35** |
| **34** | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | **34** |
| **33** | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | **33** |
| **32** | 24 | 25 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | **32** |
| **31** | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 30 | 31 | 32 | **31** |
| **30** | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | **30** |
| **29** | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | **29** |
| **28** | 21 | 22 | 23 | 24 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | **28** |
| **27** | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 28 | 29 | **27** |
| **26** | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | **26** |
| **25** | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | **25** |
| **24** | 18 | 19 | 20 | 21 | 22 | 23 | 23 | 24 | 25 | 26 | 27 | **24** |
| **23** | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | **23** |
| **22** | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | **22** |
| **21** | 16 | 17 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | **21** |
| **20** | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 21 | 22 | 23 | 24 | **20** |
| **19** | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | **19** |
| **18** | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | **18** |
| **17** | 13 | 14 | 15 | 16 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | **17** |
| **16** | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 20 | 21 | **16** |
| **15** | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | **15** |
| **14** | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | **14** |
| **13** | 10 | 11 | 12 | 13 | 14 | 14 | 15 | 16 | 17 | 18 | 19 | **13** |
| **12** | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | **12** |
| **11** | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | **11** |
| **10** | 8 | 9 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | **10** |
| **9** | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 13 | 14 | 15 | 16 | **9** |
| **8** | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | **8** |
| **7** | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | **7** |
| **6** | 5 | 6 | 7 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | **6** |
| **5** | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 12 | 13 | **5** |
| **4** | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | **4** |
| **3** | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | **3** |
| **2** | 2 | 3 | 4 | 5 | 6 | 6 | 7 | 8 | 9 | 10 | 11 | **2** |
| **1** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | **1** |

**Step 4** Determine your COMPOSITE SCORE by finding the sum of all your SCALE SCORES for each of the four sections: English only (if you do not choose to take the optional Writing Test) *or* English/Writing (if you choose to take the optional Writing Test), Math, Reading, and Science Reasoning, and divide by 4 to find the average. Round your COMPOSITE SCORE according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32.

| _____ | + | _____ | + | _____ | + | _____ | = | _____ |
|---|---|---|---|---|---|---|---|---|
| ENGLISH *OR* ENGLISH/WRITING SCALE SCORE | | MATHEMATICS SCALE SCORE | | READING SCALE SCORE | | SCIENCE SCALE SCORE | | SCALE SCORE TOTAL |

| _____ | ÷ 4 = | _____ |
|---|---|---|
| SCALE SCORE TOTAL | | COMPOSITE SCORE |

# ANSWERS AND EXPLANATIONS

## English Test Explanations

### PASSAGE I

1. **The best answer is A.** The pronoun *this* tells the reader that a description is coming further in the sentence. The sentence as it is written uses the correct verb tense and keeps the entire thought in one sentence; therefore, it's the best answer choice.

2. **The best answer is F.** The paragraph is most concerned with the writer's Grandma and her emotional relationship to the chest. Answer choices G, H, and J all shift attention away from the grandmother's personal response and towards superficial, physical descriptions.

3. **The best answer is C.** The best answer choice correctly uses commas to separate the two clauses. *Treasure* is an adjective modifying *trove*, so there is no need for a comma between them (answer choice A). Answer choices B and D include unnecessary commas after *and*.

4. **The best answer is H.** All of the answer choices have acceptable replacements for the idiomatic saying *truth be told.* However, answer choice H has the phrase *kind of type of*, which is redundant. Therefore, it is not an acceptable replacement.

5. **The best answer is A.** The first sentence sets the stage for the action, but does not give any new information. It could easily be deleted without losing the sense of the story.

6. **The best answer is G.** The object of this sentence is the *interior* of the *wooden box*. Because *wooden box* is singular and possessive, it needs an apostrophe and an s.

7. **The best answer is C.** The story is being told using the simple past tense. Answer choice A creates an incomplete sentence. Answer choice D indicates action over time, which is also not appropriate in this context.

8. **The best answer is F.** The phrase that begins with *flying* is a nonrestrictive clause. That means it could be removed without destroying the meaning of the sentence. Nonrestrictive clauses need to be set off by commas, making answer choice F best.

9. **The best answer is D.** Answer choices A and C imply a contrast that is not supported by the sentence. Answer choice B implies a conclusion that is also not supported. In fact, no connective is needed, and answer choice D is best.

10. **The best answer is G.** The underlined phrase is a list with only two items, and no commas are necessary to separate them from each other. Likewise, because it's a restrictive clause, necessary for the sentence to make sense, it doesn't need commas to separate it from the rest of the sentence.

11. **The best answer is D.** This pronoun properly refers to the writer's grandfather. Therefore answer choice D, *his*, is best.

12. **The best answer is G.** The topic of this paragraph is *dreams*—more specifically, the contrast between the dreams of the writer's grandfather and grandmother. The grandfather's dreams were to be a heroic bomber pilot, while the grandmother wanted him to do something that would keep him safe for his family. Answer choice G best introduces the tension between the two.

13. **The best answer is B.** The writer makes it clear that her grandfather took pride not only in being a pilot, but also in his wife. It would make sense that her grandfather would have a picture of the plane he named after his wife, and that the writer would refer to it as her grandfather's *pride and joy*.

14. **The best answer is F.** Use the possessive relative pronoun to refer to the writer's grandmother.

15. **The best answer is C.** This passage is intensely personal. It doesn't try to place the actions into a larger historical context, and gives very little information about the use of airplanes in World War II. Therefore, answer choices A and B are inappropriate. The passage indicates that the writer's grandparents had a loving relationship, so answer choice D is not best.

### PASSAGE II

16. **The best answer is G.** Only answer choice G has a subject (*tomatoes*), verb (*have become*), and object (*a staple*) without being in a dependent clause. The other answer choices lack a main verb and create incomplete sentences.

17. **The best answer is A.** This phrase is a three-item list; each item must be separated by a comma. Answer choices C and D are incorrect because the comma needs to be placed after *stands*, not after *and*.

18. **The best answer is F.** Answer choice G is incorrect because the phrase doesn't do anything to explain anything about the writer. The phrase also doesn't mention other summer produce, so answer choice H is incorrect. The phrase does, however, explain that tomatoes are plentiful during the growing season, which would logically lead to a need for more tomato recipes.

19. **The best answer is B.** The correct idiom is *plenty of*. Omitting the underlined portion, while not grammatically incorrect, would reduce the clarity of the sentence, which focuses on the number of ways to prepare and eat tomatoes.

20. **The best answer is F.** As written, the underlined portion introduces the idea of a contrast between recipes that are universally delicious, but have individual levels of difficulty in preparation. The remaining answer choices are either too vague or awkward.

21. **The best answer is D.** The writer wants to give a sequence of events: first peeling and deseeding, then cooking. Only answer choice D tells the reader precisely which action comes first.

22. **The best answer is F.** The verb here is in the passive voice because the action, *peeling*, is happening *to* the subject. Therefore, the correct form is *accomplished*. Because *accomplished* is a verb, it needs to be modified by the adverb, *easily*.

23. **The best answer is B.** The pronoun *whoever* is the subject of this sentence, so *who* is the correct form. Answer choices A and C use the object form *whom*, and should be eliminated. The passage is written in the present tense, so answer choice B is correct.

24. **The best answer is H.** In this sentence, the phrase surrounding *cool enough to handle* doesn't have a main verb, so it can't stand on its own. This eliminates all choices except answer choice H.

25. **The best answer is D.** The paragraph explains in great detail how to peel and deseed tomatoes. The last sentence tells us this is important, but only answer choice D includes information that tells us why. Answer choices C and D explain why it might be important to use a strainer, but their scope doesn't cover the entire paragraph. Answer choice B introduces a new idea, growing new plants, that is also outside the scope of the paragraph.

26. **The best answer is H.** This sentence is actually two complete sentences. Joining them with a comma, as in answer choice F, is called a "comma splice" and is grammatically incorrect. Answer choice G creates a run-on sentence. Answer choice J implies a contrast that doesn't make sense in this context.

27. **The best answer is A.** The adjective *bright* modifies *red*, so the words don't need to be separated by a comma. This eliminates answer choices B and D. However, the phrase *Once a tomato...* is a nonrestrictive relative clause and does need to be separated from the main sentence by a comma.

28. **The best answer is J.** The phrase doesn't add significant details or explain why fresh vegetables are important. Therefore, answer choices F and G can be eliminated. Both of the remaining phrases are specific, so answer choice H can be eliminated. Answer choice J is best because the proposed replacement is much wordier than the original without adding anything significant.

29. **The best answer is D.** The pronoun here refers back to the noun *Italians*. Therefore, the plural pronoun, *them*, is best.

30. **The best answer is H.** The key phrase in this question is *modern cuisine*. The essay's focus is much more general than how tomatoes are used in a particular type of cuisine. Therefore, answer choices F and G can be eliminated. The essay does mention specific recipes, so answer choice J can be eliminated. Answer choice H is best because the writer takes a general approach and includes pieces of information like the fact that Italians once thought that tomatoes were poisonous.

## PASSAGE III

31. **The best answer is B.** By definition, a *hole* is empty, so the underlined portion, as it is, is somewhat redundant. The adjective that best fits the tone of the paragraph is *gaping*. Omitting the underlined portion eliminates the modifier *a*, creating an awkward, ungrammatical sentence.

32. **The best answer is F.** The phrase beginning with *gumming* is a nonrestrictive relative clause; therefore it needs to be set apart from the main sentence by a comma. The phrase is not a complete sentence, so answer choices G and H should be eliminated.

33. **The best answer is A.** This paragraph introduces a new idea: the tools and techniques of modern dentistry. No connective to the previous paragraph is necessary, so answer choice A is best. Answer choices C and D are incorrect because they imply continuity with the previous paragraph. Answer choice B implies a level of contrast that isn't necessary.

**34. The best answer is J.** Answer choices F, G, and H are all redundant. By definition, a disposable needle is one that can be disposed of. Answer choice J has the all the information needed; therefore, it's the best answer.

**35. The best answer is A.** The focus of this paragraph is equipment; therefore, answer choice A is best because it describes the widespread use of autoclaves in modern dentistry. Answer choice B should be eliminated because the phrase does not further define the term *autoclave*. Answer choice C is incorrect because the phrase helps explain *how* the offices had become more sterile. Answer choice D can be eliminated because the focus of the paragraph is not on teeth.

**36. The best answer is J.** The sentence already tells the reader that the equipment uses heat to sterilize the equipment. Therefore, the phrase beginning *by using the heat* is redundant.

**37. The best answer is B.** The verb *work* has to be in the present tense to match the rest of the paragraph. Therefore, answer choices A and C can be eliminated. Answer choice D would need the auxiliary verb *are* to be correct.

**38. The best answer is H.** The contrast *on the other hand* needs to be separated from the main sentence by a comma. The phrase, *constant changes*, however, is the subject of the main clause and doesn't need a comma. This phrase is also followed by a restrictive relative clause, which requires no comma.

**39. The best answer is A.** Answer choices C and D imply a contrast with the previous sentence, and so can be eliminated. Answer choice B makes the phrase a separate clause. Answer choice A is correct because the phrase is part of a noun phrase that acts as the subject of the sentence and doesn't need a conjunction.

**40. The best answer is G.** All of the answer choices offer some added detail. However, answer choices F, H, and J are too vague. Only answer choice G gives a detail that improves the reader's understanding of the challenges faced by modern dentists as opposed to those of 50 years ago.

**41. The best answer is A.** This sentence is really two complete sentences, and therefore needs to be separated by a period or a semicolon. Answer choice B creates a comma splice, answer choice C creates a run-on sentence, and answer choice D creates a long, incomplete sentence.

**42. The best answer is H.** Answer choices F, G, and J are either too wordy or grammatically incorrect. Answer choice H is both succinct and grammatically correct, so it is best.

**43. The best answer is D.** The phrase *after 1950* means the same thing as *second half of the twentieth century*. So, answer choices A, B, and C are all redundant. Answer choice D is best because it has all the information necessary without repeating itself.

**44. The best answer is F.** The tone of the passage is scientific and dispassionate. Answer choice G, therefore, should be eliminated because it has the wrong tone. Answer choice H is too vague, and answer choice J is off topic. Answer choice F explains precise ways in which dentistry has improved over the past 50 years, which relates directly back to the main topic of the essay.

**45. The best answer is D.** Paragraph 3 does not contain detailed information about dental practices, as do Paragraphs 2 and 4. Instead, it's more of a summary of the changes in dentistry over time. Therefore, the best place for it is at the end of the essay, which makes answer choice D best.

## PASSAGE IV

**46. The best answer is J.** Here, *Mr. Peale* is a type of nonrestrictive clause modifying *our Advanced English teacher*, and should be separated by commas from the rest of the sentence.

**47. The best answer is A.** *That we especially liked* is a restrictive relative clause. That means, if it were removed, the basic sense of the sentence would change. Restrictive relative clauses do not take commas, so answer choices B and D can be eliminated. The final clause is not independent, so it does not take a comma. This eliminates choice C.

**48. The best answer is G.** Because the list of painters is the object of the verb *shared*, a colon is incorrect; eliminate answer choice F. Commas are not used to begin lists, so eliminate answer choice H. Answer Choice J, *share*, is in the wrong tense.

**49. The best answer is D.** The topic of this paragraph is to introduce a day when the writer had to reproduce a famous painting in her English class. Answer choices A and C provide specific details that are not followed up in the next paragraph. Therefore, they are off topic and should be eliminated. Answer choice B does not have the right tone and is also off topic. The best choice is to delete the sentence altogether because it is irrelevant.

**50. The best answer is H.** Answer choices F, G, and J are redundant. Answer choice H, however, contains all the necessary information without any extra words.

**51. The best answer is B.** The passage has been written in the past tense. Therefore, answer choices A and C can be eliminated. Answer choice D doesn't make grammatical sense, so it should be eliminated.

**52. The best answer is J.** The word *proclaimed* means to state loudly and with assurance. Answer choices F, G, and H are all close synonyms. Answer choice J, however, indicates that Mr. Peale put his thoughts in writing, which is not an acceptable alternative.

**53. The best answer is C.** The writer's phrase is not directed at either her teacher or Impressionist art. So, answer choices A and B can be eliminated. She is instead commenting on how his response made her feel—which was ambivalent.

**54. The best answer is F.** The subject of Sentence 3 is Mr. Peale's lack of comment. It would not make sense to place the sentence before Sentence 2 because that's where the writer gives that information. So, eliminate answer choices G and H. By Sentence 4, Mr. Peale has moved on, so eliminate answer choice J. Answer choice F makes the most logical sense.

**55. The best answer is C.** Answer choices A and B are unnecessarily wordy. Answer choice D implies a relative clause that isn't there. Answer choice C is grammatically correct and concise.

**56. The best answer is F.** The proposed sentence needs to link the idea that the writer had captured the essence of the original painting and that she valued her reproduction enough to hang it on her wall, even though it wasn't terribly dramatic. The proposed sentence does this by emphasizing that the reproduction had captured the parts of the original most valued by the writer: its beauty and serenity.

**57. The best answer is B.** The correct sentence needs to be in the past tense, to stay consistent with the rest of the essay and use of active voice. So, answer choice A can be eliminated. Answer choice C doesn't make grammatical sense. Answer choice D has the awkward and unnecessary phrase *by me*, so it also can be eliminated.

**58. The best answer is G.** Answer choice F implies a conditional state not supported by the sentence. Answer choice H implies a contrast not supported by the sentence. Answer choice J does not make sense. Only answer choice G provides an appropriate adverb.

**59. The best answer is C.** Answer choices A and B are wordier versions of answer choice C, so they can be eliminated. Answer choice D doesn't make sense and can be eliminated.

**60. The best answer is J.** Answer choice F is in the future tense, so eliminate it. Answer choices G and H are awkward and wordy, so eliminate them. Answer choice J is concise and grammatically correct.

### PASSAGE V

**61. The best answer is A.** The conjunction *however* needs to be separated from the rest of the sentence by commas. Answer choices B and C create incomplete sentences.

**62. The best answer is G.** The phrase *in Africa* is a restrictive clause that should not be separated from the rest of the sentence by punctuation.

**63. The best answer is D.** The use of the modifier *other* instead of *another* tells the reader that the noun *countries* needs to be plural. Additionally, the introductory clause must be separated by a comma from the main clause. Answer choice C uses a comma correctly, but it incorrectly includes the plural possessive form of the noun.

**64. The best answer is H.** The conjunction here needs to connect this sentence to the previous one by suggesting that this sentence offers an example. Answer choices F and G imply a contrast to the previous sentence. Answer choice J implies a comparative. Only answer choice H correctly implies that the tall trees on plantations around the world assisted in the growth of coffee plants by simulating their preferred habitats.

**65. The best answer is B.** The two phrases are each complete sentences and must be joined by either a semicolon or some form of end punctuation. Answer choice C creates a "comma splice" by using a comma to join two independent clauses.

**66. The best answer is H.** This paragraph is written in the present tense. Therefore, the simple past tense is correct. Answer choice H is also in the active voice, making it the best choice.

**67. The best answer is C.** The sense of the phrase is that the new plants are sun-tolerant *and also* able to thrive in open sunlight. All of the choices have this sense of continuity except answer choice C.

**68. The best answer is J.** The phrase *amount of* is idiomatic and should be used here. The remaining answer choices are not idiomatic.

**69. The best answer is A.** The conjunction here needs to give the sense of result or conclusion. Answer choices B and C imply contrast. Answer choice D implies continuity, but not result.

**70. The best answer is G.** The key to this question is *effective visual detail*. Answer choice G offers the most precise detail because it specifies that the ground protection is from *trees* and that the rainfall is *pounding*.

**71. The best answer is C.** Because *ornithologists* is the subject of the sentence and *discover* is the verb, the relative pronoun *who* is unnecessary. Therefore, answer choices A and B can be eliminated. Answer choice D is in the past tense, while the rest of the paragraph is in the present tense.

**72. The best answer is J.** The conjunction *as* implies a reason or explanation, or even action over time. Consequently, answer choices F, G, and H would all be acceptable alternatives. Answer choice J, *therefore*, implies a conclusion or direct result. It is not an acceptable alternative to *as*.

**73. The best answer is A.** The phrase *as well as many other fauna and flora* is a nonrestrictive clause because removing it does not significantly alter the meaning of the sentence. A nonrestrictive clause is separated from the main sentence by commas.

**74. The best answer is J.** Sentence 3 tells the reader about recorded decreases in the songbird population of the northern United States. However, the effect of coffee production on birds is not introduced until Sentence 5. Therefore, Sentence 3 should logically come after Sentence 5, after Sentence 6 most specifically. That makes answer choice J, *before Sentence 7*, the best answer.

**75. The best answer is A.** Answer choice A clearly and concisely states the idea of many people coming to a realization. The other choices state the same idea, but they are grammatically awkward and too wordy or redundant.

## Mathematics Test Explanations

1. **The correct answer is C.** To solve this problem, multiply the number of ounces by the number of pounds:

$$16 \times 1.5 = 24$$

2. **The correct answer is J.** To solve this problem, use the FOIL method then combine like terms, as follows:

First terms: $3x^2$

Outside terms: $-15x$

Inside terms: $4x$

Last terms: $-20$

$$3x^2 - 15x + 4x - 20 = 3x^2 - 11x - 20$$

3. **The correct answer is B.** To solve this problem, substitute 8 for $a$ and 9 for $b$ in the given function and solve, as follows:

$$\begin{aligned} f(a,b) &= ab - (a-b) \\ f(8,9) &= (8)(9) - (8-9) \\ &= 72 - (-1) \\ &= 72 + 1 = 73 \end{aligned}$$

4. **The correct answer is G.** To solve this problem, first calculate 16% of 24,000, as follows:

$$24{,}000 \times .16 = 3{,}840$$

Next, calculate 1/5 of 3,840, as follows:

$$\begin{aligned} &3{,}840 \times 1/5 \\ &= 3{,}840 \div 5 = 768 \end{aligned}$$

5. **The correct answer is B.** Solve this problem by isolating $x$ on the left side of the equation. Perform the following operations:

$$5x + 5 = 25 + 3x$$

Subtract $3x$ from both sides:

$$2x + 5 = 25$$

Subtract 5 from both sides:

$$2x = 20$$

Divide both sides by 2:

$$x = 10$$

6. **The correct answer is G.** Remember that the perimeter is the distance around an object. In this case, because the figure is a parallelogram, its perimeter is twice the sum of the lengths of two adjacent sides. You are given that the length of one side, $AB$ is 7 inches. Set up the following equation with $x$ as the unknown side length, $AD$:

$$\begin{aligned} 46 &= 2(7 + x) \\ 46 &= 14 + 2x \\ 32 &= 2x \\ 16 &= x \end{aligned}$$

7. **The correct answer is E.** To solve this problem, recall that absolute values are always positive. Therefore, you can ignore the negative sign in front of 2 and solve, as follows:

$$\begin{aligned} &|14| \times |-2| \\ &= 14 \times 2 = 28 \end{aligned}$$

8. **The correct answer is J.** To solve this problem you should notice that, because line $XY$ is parallel to line $AB$, a new triangle, $XYC$, which is similar to triangle $ABC$ has been created. Therefore, the corresponding angle measures in each triangle will be congruent, which means that angle $A = 40°$. You are given that the measure of angle $B$ is 80°. The sum of the measure of angles in a triangle is 180°, so set up the following equation to solve for the measure of angle $C$:

$$\begin{aligned} 180° &= 40° + 80° + C \\ 180° &= 120° + C \\ 60° &= C \end{aligned}$$

9. **The correct answer is E.** To solve this problem, simply substitute $-3$ for $x$ and 2 for $y$ in the equation, as follows (carefully track the negative sign!):

$$\begin{aligned} x^3y + xy^3 &= (-3)^3(2) + (-3)(2)^3 \\ &= (-27)(2) + (-3)(8) \\ &= -54 - 24 \\ &= -78 \end{aligned}$$

10. **The correct answer is F.** One way to solve this problem is to draw a chart like the one shown below. You are given that $y$ represents the number of full years of employment. Therefore, the salary

increase for each professor should be shown as a coefficient of $y$:

| A | B |
|---|---|
| 50,000 + 1,500$y$ | 42,000 + 2,800$y$ |

Now that you know how to calculate each professor's salary after $y$ number of years, set these values equal to each other to come up with the correct equation:

$$50{,}000 + 1{,}500y = 42{,}000 + 2{,}800y$$

**11. The correct answer is B.** Don't let the fact that there are no numbers in this math problem confuse you! Simply remember that $W$, $X$, $Y$, and $Z$ each represent some number, and perform the correct mathematical operations to isolate $Z$ on one side of the equation, as follows:

$$W = XYZ$$
$$\frac{W}{XY} = Z$$

**12. The correct answer is K.** You are given that both numbers have a factor of 8 and that they both factor into 48 evenly (48 is the least common multiple). Therefore, the following is true:

$$48 = 8 \times a \times b$$

Because $48 = 8 \times 6$, $a \times b$ must equal 6; $a$ could equal 2 and $b$ could equal 3, which means that one of the given numbers has a factor of 2 and the other has a factor of 3. Both numbers have a common factor of 8, so one number could be $8 \times 2 = 16$ and the other number could be $8 \times 3 = 24$.

**13. The correct answer is B**. To solve this problem, set up an equation to determine the length of the shorter piece, substituting $x$ for the unknown, shorter length:

$$x + (x + 23) = 55$$
$$2x + 23 = 55$$
$$2x = 32$$
$$x = 16$$

The problem states that one piece is 23 feet longer than the other piece, which does not necessarily mean that the shorter piece will be 23 feet long, so answer choice A is incorrect. If you simply subtracted 23 from 55 you would get answer choice D, which is incorrect. This calculation does not account for both lengths of rope. If you divided the original length of the rope (55) by 2, subtracted 23 and rounded up, you would get answer choice E. This is not the correct calculation to perform.

**14. The correct answer is F.** To solve this problem, expand $(x + r)^2$ using the FOIL method:

$$(x + r)^2$$
$$= (x + r)(x + r)$$
$$= x^2 + 2xr + r^2$$

Now, substitute this value for $(x + r)^2$ in the given equation:

$$x^2 + 2xr + r^2 = x^2 + 22x + r^2$$

You can see that $2xr = 22x$, so solve for $r$, as follows:

$$2xr = 22x$$
$$r = 22x/2x$$
$$r = 11$$

**15. The correct answer is C.** To solve this problem, first find the common denominator, as follows:

$$3\frac{1}{3} + 2\frac{4}{5} = 3\frac{5}{15} + 2\frac{12}{15}$$

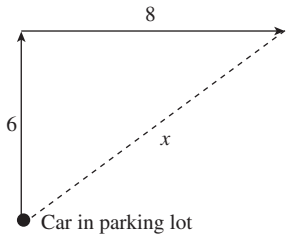
Next, add the fractions:

$$3\frac{5}{15} + 2\frac{12}{15}$$
$$= 5\frac{17}{15}$$
$$= 6\frac{2}{15}$$

Now look at the fraction part of this mixed number and work through the answer choices. You know that $\frac{2}{15}$ is less than $\frac{1}{2}$ $(\frac{7.5}{15})$, so eliminate answer choice A. You also know that $\frac{2}{15}$ is less than $\frac{1}{3}$ $(\frac{5}{15})$, so eliminate answer choice B. Because $6\frac{2}{15}$ is greater than 6 but less than

$6\frac{1}{3}$, Jenny's total distance is between 6 and $6\frac{1}{3}$ miles.

**16. The correct answer is H.** To solve this problem, it might be helpful to draw a picture like the one shown next:




You can see that the route traveled creates a right triangle. Use the Pythagorean Theorem to calculate the distance from the parking lot, as follows:

$$6^2 + 8^2 = x^2$$
$$36 + 64 = x^2$$
$$100 = x^2$$
$$10 = x$$

**17. The correct answer is B.** A line, parabola, and so on will intersect the $x$-axis when $y = 0$. Therefore, you can set each of the elements given in the question to 0 and solve for $x$, as follows:

$$(x + 1) = 0, x = -1$$
$$(x - 2) = 0, x = 2$$
$$(x + 3) = 0, x = -3$$
$$(x + 4) = 0, x = -4$$

The graph will cross the $x$-axis at $-1, 2, -3$, and $-4$.

**18. The correct answer is G.** To solve this problem, first calculate the new area of Marcia's horse corral:

$$\text{Initial area} = 50 \times 125 = 6{,}250 \text{ square feet}$$
$$\text{New area} = 6{,}250 + 1{,}850 = 8{,}100 \text{ square feet}$$

Now, you can try the answer choices to see which combination of width and length fits the parameters given in the question:

**Answer choice F:** While these dimensions indicate an increase of 5 feet for both the width and the length, the area does not equal 8,100 ($55 \times 130 = 7{,}150$ square feet).

**Answer choice G:** These dimensions reflect an increase of 10 feet for both the width and the length, and the area equals 8,100 ($60 \times 135 = 8{,}100$ square feet), so this is the correct answer.

**19. The correct answer is D.** To solve this problem, first remember that perimeter is equal to the distance around an object. Therefore, the perimeter of a triangle is simply the sum of the lengths of its sides. You are given that the lengths of the sides of the triangle are 3 consecutive even integers. Set the first length (the shortest side) equal to $x$, the second length equal to $x + 2$, and the third length (the longest side) equal to $x + 4$. Now, create an equation and solve for $x$, as follows:

$$x + (x + 2) + (x + 4) = 48$$
$$3x + 6 = 48$$
$$x = 14$$

You now know that the length of the shortest side is 14, so the length of the longest side must be $14 + 4 = 18$.

**20. The correct answer is H.** To solve this problem, put the given equation in the standard form, $y = mx + b$, where $m$ is the slope:

$$4y - 6x = 8$$
$$4y = 6x + 8$$
$$y = \left(\frac{6}{4}\right)x + 2$$
$$y = \left(\frac{3}{2}\right)x + 2$$

Therefore, $m = \frac{3}{2}$.

**21. The correct answer is B.** One way to solve this problem is to use SOH CAH TOA to find the sin, cos, and tan of angle $X$, as follows:

$$\text{SOH: sin} = \text{opposite/hypotenuse} = \frac{12}{13}$$
$$\text{CAH: cos} = \text{adjacent/hypotenuse} = \frac{5}{13}$$
$$\text{TOA: tan} = \text{opposite/adjacent} = \frac{12}{5}$$

Only answer choice B, $\sin X = \frac{12}{13}$, is a true statement about angle $X$.

**22. The correct answer is F.** To solve this problem, draw a line from $Q$ to $S$ on the figure, creating right

triangle $QST$. You are given that the length of $QT$ is 5 and the length of $ST$ is 3. Use the Pythagorean theorem to find the length of $QS$, the hypotenuse:

$$3^2 + 5^2 = c^2$$
$$9 + 25 = c^2$$
$$34 = c^2$$
$$\sqrt{34} = c$$

**23. The correct answer is E.** To solve this problem, recall that a counterclockwise rotation will move point $R$ to the left about the origin. Therefore, the $x$-value will be negative, and you can eliminate answer choices A, B, and C. Likewise, a rotation about the origin will move point $R$ from its current location, so the $x$-value will change, leaving answer choice E as the only option.

**24. The correct answer is H.** Recall that the perimeter is the distance around any object. You are given the side lengths, so start by finding the sum of the given lengths:

$$QT = 5, RS = \sqrt{34}, ST = 3, \text{and } RQ = 6$$
$$5 + \sqrt{34} + 3 + 6$$
$$= 14 + \sqrt{34}$$

At this point you can safely eliminate answer choices J and K because they are too small. You are asked for an approximation, so you don't actually have to do all the math. The square root of 34 is slightly less than the square root of 36, so the perimeter will be slightly less than $14 + 6 = 20$, leaving you with 19.8 as the only viable option.

**25. The correct answer is B.** To solve this problem, first convert the information in the question into its mathematical equivalent, as follows:

5 times a number $x = 5x$

Subtracting $5x$ from 15 yields a negative result, so

$15 - 5x < 0$.

Now, solve the inequality for x:

$$15 - 5x < 0$$
$$-5x < -15$$

Divide both sides by $-5$ and reverse the inequality to get $x > 3$.

**26. The correct answer is K.** One way to solve this problem is to substitute each of the answer choices for $t$ in the inequality to determine which one does NOT work:

**Answer choice F:** $|74 - 34| = |40| = 40$; $40 \leq 40$, so eliminate answer choice F.

**Answer choice G:** $|16 - 34| = |-18| = 18$; $18 \leq 40$, so eliminate answer choice G.

**Answer choice H:** $|0 - 34| = |-34| = 34$; $34 \leq 40$, so eliminate answer choice H.

**Answer choice J:** $|-6 - 34| = |-40| = 40$; $40 \leq 40$, so eliminate answer choice J.

By the process of elimination, answer choice K must be correct: $|-8 - 34| = |-42| = 42$, which is NOT $\leq 40$.

**27. The correct answer is D.** The slope-intercept form of the equation for a line is $y = mx + b$; isolate y on the left side of the equation, as follows:

$$10x - y - 8 = 0$$
$$10x - y = 8$$
$$-y = -10x + 8$$
$$y = 10x - 8$$

**28. The correct answer is H.** As shown in the figure, the radius, $r$, is the hypotenuse of a right triangle. One leg of the triangle is 6 and the other is half of 32, or 16. Use the Pythagorean theorem to calculate the length of $r$, as follows:

$$6^2 + 16^2 = r^2$$
$$36 + 256 = r^2$$
$$292 = r^2$$
$$17.08 = r$$

Rounded to the nearest tenth of a centimeter, $r = 17.1$ cm.

**29. The correct answer is B.** Each of the numbers in the values given must divide evenly into the least common multiple. Therefore, you can quickly eliminate answer choice A because 4 does not divide evenly into 15. Next, notice that each of the number values divides evenly into 60, which is less than 120, so eliminate answer choices D and E. Because $ab$ is a smaller multiple than $a^2b$, and $ab$ divides evenly into the product of all of the given values, answer choice B is the least common multiple of the given values.

**30. The correct answer is K.** You are given that the two lines intersect at a certain point, which means that the lines cross at the *same* point in the $(x,y)$ coordinate plane, so at that point, the equations of those lines is equal. Set up the following equation and solve for $x$:

$$3x + 4 = 2x + 6$$
$$x = 2$$

Now, substitute 2 for $x$ in either equation and find $y$, as follows:

$$y = 3x + 4$$
$$y = 3(2) + 4$$
$$y = 6 + 4 = 10$$

**31. The correct answer is E.** To solve this problem, first expand the numerator and factor the denominator, then simplify, as follows:

$$\frac{(x-13)(x-13)}{(x-13)(x+13)} = \frac{(x-13)}{(x+13)}$$

**32. The correct answer is F.** To solve this problem, first recognize that, if $n = m + 2$, then $m - n = -2$. Now, substitute this value into the second expression and solve:

$$(m-n)^4 = (-2)^4 = 16$$

**33. The correct answer is D.** To solve this problem, first convert the information given into its mathematical equivalent, as follows (use $b$ to represent the larger number):

The larger of two numbers exceeds twice the smaller number by nine: $b = 2a + 9$

The sum of twice the larger and five times the smaller number is 74: $2b + 5a = 74$

Now, simply substitute the value of $b$ into the second equation in order to solve for $a$:

$$2(2a + 9) + 5a = 74$$

**34. The correct answer is G.** You are given that $x/y = 4$, so $x = 4y$. Substitute $4y$ for $x$ in the equation and solve, as follows:

$$x^2 - 12y^2$$
$$= (4y)^2 - 12y^2$$
$$= 16y^2 - 12y^2$$
$$= 4y^2$$

**35. The correct answer is B.** You are given certain ratios in this problem, so you can set up a proportion to find the length of the shortest side of the second triangle. The ratio of the longest sides of both triangles is $\frac{15}{10}$. Set the length of the shortest side of the second triangle to $x$ so that the ratio of the shortest sides of both triangles is $\frac{12}{x}$. Now, set up a proportion, cross-multiply, and solve for $x$:

$$\frac{15}{10} = \frac{12}{x}$$
$$15x = 120$$
$$x = 8$$

**36. The correct answer is F.** One way to solve this problem is to recognize that $ab^2$ is a common factor of $a^2b^2$ and $ab^3$. Therefore, it must also be a factor of 45, which can be factored as $5 \times 3^2$. Now you can see that $b$ could be 3, and since there can only be one correct answer, F must be it. If you test $a = 5$ and $b = 3$, you see that the greatest common factor of $5^2 \times 3^2$ and $5 \times 3^3$ is $5 \times 3^2$, which equals 45.

**37. The correct answer is C.** To solve this problem, set the fixed charge to $x$ and set the number of half-miles to $y$. Now, select two values from the table to create a set of equations, and solve for $x$ and $y$, as follows:

$$x + 5y = 8.00$$
$$x + 6y = 8.50$$

The $x$ values will cancel out, leaving $y = 0.50$, which means that the cost per half-mile is \$0.50; eliminate answer choice A. Now, substitute .5 for $y$ in one of the equations to solve for $x$:

$$x + 5y = 8$$
$$x + 5(.5) = 8$$
$$x = 8 - 2.5 = 5.50$$

**38. The correct answer is G.** One way to solve this problem is to use SOH CAH TOA for angle $A$. You are given that the sine of angle $A$ is 3/5, which means that the ratio of the length of the side opposite angle $A$ to the length of the hypotenuse is 3 to 5. So, the ratio 3/5 is equal to the ratio $BC/AC$. You are given that $AC$, the hypotenuse, is equal to 16, so now you can set up a proportion, cross-multiply,

and solve for $BC$, as follows:

$$3/5 = BC/AC$$
$$5BC = 3AC$$
$$5BC = 3(16)$$
$$5BC = 48$$
$$BC = 9.6$$

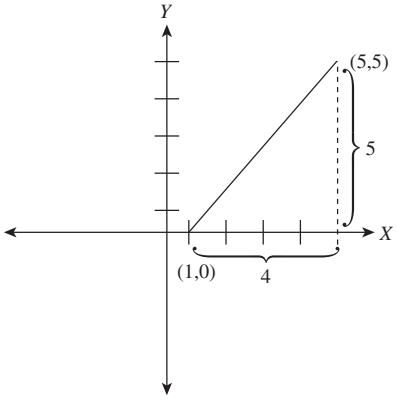
**39. The correct answer is E.** The equation of a circle is $(x-h)^2 + (y-k)^2 = r^2$, where $h$ and $k$ are the $x$ and $y$-coordinates of the center of the circle and $r$ is the radius. Therefore, the equation of the circle centered at point $(12, -5)$ with a radius of 4 is $(x-12)^2 + (y+5)^2 = 16$.

**40. The correct answer is G.** One way to solve this problem is to recognize that, in order for $\frac{24}{30}$ to be greater than $\frac{t}{24}$, $t$ must be less than 24. This is true because any fraction with a denominator larger than or equal to its numerator will always be *less* than a fraction whose numerator is larger than or equal to its denominator. So, $\frac{t}{24}$ is only less than $\frac{24}{30}$ when $t$ is less than 24. Because you are asked for the largest integer value of $t$ and you know that $t$ must be less than 24, the correct answer must be 19. You could also set the values equal to each other, cross-multiply and solve for $t$, as follows:

$$\frac{24}{30} = \frac{t}{24}$$
$$30t = 576$$
$$t = 19.2$$

The integer value that makes $\frac{24}{30}$ greater than $\frac{t}{24}$, therefore, is 19.

**41. The correct answer is B.** To solve this problem you could either use the distance formula, or sketch a quick figure like the one shown below:

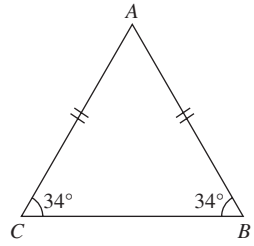



You can see that the distance between the points is the hypotenuse of the right triangle with side lengths of 5 and 4. Use the Pythagorean theorem to solve for the hypotenuse, $c$:

$$c^2 = 5^2 + 4^2 = 41$$
$$c = \sqrt{41}$$

**42. The correct answer is K.** To solve this problem, recognize that $MN$ is parallel to $LO$ and that these lines are cut by the transversal $MO$. Therefore, angles $OMN$ and $MOL$ are alternate interior angles and they have the same measure, 30°. Now, because a straight line has 180°, the measure of angle $MON$ must be $180° - 30° - 105° = 45°$.

**43. The correct answer is E.** A good way to solve this problem is to sketch triangle $ABC$, as shown below:




You are given that $AB$ is congruent to $AC$, and that the measure of angle $B$ is 34°. This means that the measure of angle $C$ is also 34°, and the measure of angle $A$ is $180° - 34° - 34° = 112°$.

**44. The correct answer is K.** To solve this problem, first set the total amount of the budget equal to $b$. Housing costs are equal to 30 percent of $b$, or $.3b$. Of this portion, 20 percent, or .2, goes toward rent. Therefore, rent, $r$, is equal to $.2(.3b)$, or $.06b$. You are given that rent, $r$, equals \$630, so now you can set up an equation and solve for $b$, as follows:

$$.06b = 630$$
$$b = 630/.06$$
$$b = 10,500$$

**45. The correct answer is A.** To find the matrix product, simply multiply the terms from a row in the first matrix by the corresponding term

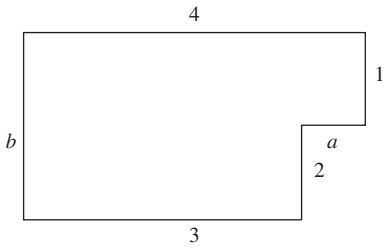
from a column in the second matrix, as shown next:

$$2x \times 1 = \mathbf{2x} \quad 2x \times 0 = \mathbf{0} \quad 2x \times -1 = \mathbf{-2x}$$

$$3x \times 1 = \mathbf{3x} \quad 3x \times 0 = \mathbf{0} \quad 3x \times -1 = \mathbf{-3x}$$

$$5x \times 1 = \mathbf{5x} \quad 2x \times 0 = \mathbf{0} \quad 5x \times -1 = \mathbf{-5x}$$

**46. The correct answer is G.** To solve this problem, you will need to find the two missing dimensions. Sketch the figure as shown below to help visualize the missing dimensions:




You can see that $a$ must be 1 and $b$ must be 3. Therefore, the perimeter is $4+3+2+1+1+3 = 14$.

**47. The correct answer is A.** To solve this problem, first recognize that $\frac{a+b}{2} = 6$, which means that $a + b = 12$. Also, $\frac{a+b+c}{3} = 11$, which means that $a + b + c = 33$. Therefore, $12 + c = 33$, so $c$ must equal $33 - 12$, or 21.

**48. The correct answer is F.** One quick way to solve this problem is to start with 12 and add each preceding even integer until you get to a sum of 40, and then count the terms:

$$12 + 10 + 8 + 6 + 4 = 40.$$

There are 5 terms in the set.

**49. The correct answer is B.** The absolute value of any number is positive. Eliminate answer choice A because it is negative. Since $w$ is close to 1.75 on the number line shown, and $u$ is close to 0.6 on the number line shown, the difference of $w$ and $u$ is close to 1.15 on the number line shown. This corresponds to $v$.

**50. The correct answer is G.** To solve this problem, you must remember that the formula for the circumference of a circle is $C = 2\pi r$, where $r$ is the radius. You are given that the length of arc $XY$ is $7\pi$, and that this is 1/6 of the total circumference. Therefore, the circumference must be $6 \times 7\pi = 42\pi$. Substitute this value for C in the equation and solve for $r$, as follows:

$$42\pi = 2\pi r$$

$$21\pi = \pi r$$

$$21 = r$$

**51. The correct answer is E.** To solve this problem, start with $n = 1$ until you reach an answer choice: when you reach $n = 8$, you will get $2(8)^2 - 6(8) = 80$, answer choice E.

**52. The correct answer is K.** To solve this problem, first find the values of $x$ for which $g(x) = 69$ by solving the equation $69 = 3(x^2 - 2)$, as follows:

$$69 = 3(x^2 - 2)$$

$$23 = x^2 - 2$$

$$25 = x^2$$

$$x = 5 \text{ or } x = -5$$

The possible values of $2x - 3$ are therefore 7 or $-13$, of which 7 is the only available answer choice.

**53. The correct answer is B.** The correct answer will be the statement that is always true. Because $a$ is greater than 4, and $25 - 4 = 21$, $b$ must always be less than 21.

**54. The correct answer is J.** You are given that $m + n$ is even and the value of $(m + n)^2$ is also even. However, because $(m + n)^2 + n + p$ is odd, the sum $n + p$ must be odd. A sum of two positive integers is odd only when one is even and one is odd. Therefore, it must be true that if $n$ is even, $p$ is odd.

**55. The correct answer is B.** Since you cannot have a partial coin, the total number of coins in the bag must be divisible by both 6 and 4 (1/6 are quarters and 1/4 are nickels). The only answer choice that is divisible by both 6 and 4 is 24.

**56. The correct answer is H.** To solve this problem, you need to recognize that the slope of the line

is equal to 4. In the standard equation for a line, $y = mx + b$, $m$ is equivalent to the slope. The slope is equal to the change in $y$-values over the change in $x$-values; set up the following equation to solve for $t$:

$$\text{slope} = (s + t) - s/(r + 2) - r$$
$$4 = t/2$$
$$8 = t$$

**57. The correct answer is E.** You are given that $x > 90$, which means that $y$ must be less than 90, because it is a complimentary angle to $x$. The sum of complimentary angles is 180. None of the other answer choices must necessarily be true.

**58. The correct answer is K.** You are given that $x = 2y + 3z - 5$. If the value of $y$ decreases by 1, then the new value of $y$ is $y - 1$; if the value of $z$ increases by 2, then the new value of $z$ is $z + 2$. Substitute these values into the original equation to see the effects of the changes on the value of $x$:

$$x = 2(y - 1) + 3(z + 2) - 5$$
$$x = 2y - 2 + 3z + 6 - 5 = 2y + 3z - 1$$

Therefore, $x$ now equals $2y + 3z - 1$, which is 4 more than $2y + 3z - 5$.

**59. The correct answer is E**. The distance between $R$ and $N$ is the length of a diagonal of the cube, whereas the distance from $S$ to $N$ is the length of a side of the cube. The length of the diagonal is longer than the length of the side. Therefore, the distance from $N$ to $R$ is not the same as the distance as $N$ to $S$.

**60. The correct answer is F.** Since $t$ is the supplement of the 35 degree angle, $t = 180 - 35$, or 145°. Because $s$ is the supplement to $t$, $s = 180 - 145$, or 35°. Since $p$ is the complement of $s$, $p = 90 - 35$, or 55°. Therefore, $3p + 4s - 2t = 3(55) + 4(35) - 2(145) = 165 + 140 - 290 = 15°$.

## Reading Test Explanations

### PASSAGE I

1. **The best answer is B.** The tone of the first seven paragraphs suggests that Libby is not as excited about the move as is her husband John. The passages states, "… hoping the enthusiasm in my voice would match the sparkle in his eyes" to indicate that Libby was attempting to appear excited for John's sake. The other answer choices are clearly indicated in the first seven paragraphs.

2. **The best answer is J.** The passage states that, "The fall semester was just a few short weeks away." This statement and the mention elsewhere in the passage of the hot weather indicates that it was most likely late summer when the couple moved into their new apartment.

3. **The best answer is A.** According to the passage, Libby initially was less enthusiastic than John about the move, but eventually changed her attitude. The passages states, "… hoping the enthusiasm in my voice would match the sparkle in his eyes" to indicate that Libby was initially attempting to appear excited for John's sake. This information best supports answer choice A.

4. **The best answer is H.** According to the passage, Libby initially was less enthusiastic than John about the move, but eventually changed her attitude. The passage states that, "There's something monumental about your first step into your new home … Now, suddenly, I knew the future had begun … I knew it could be anything we wanted it to be." These statements best support an attitude of cautious optimism. While Libby was not very excited when she first arrived, nothing in the passage suggests that she dreaded the move, so eliminate answer choice G. Likewise, the other answer choices are not supported by details in the passage.

5. **The best answer is A.** The passage states that, "We each grabbed an armful and headed towards our new apartment, just a few blocks away from the squat, industrial-looking building where John would start teaching." This statement best supports answer choice A. While Libby was accustomed to university buildings that were ivy-covered, John's new workplace was different.

6. **The best answer is F.** The passage does not describe the hardships faced by a young couple in their first home. While there is some mention of Libby's initial reservations about moving into the new apartment, there is not enough detail about any hardships she and John may have faced to support answer choice F as an accurate description of the passage.

7. **The best answer is C.** The question asks about John's perception of his wife. When John says, "Libby, we have an entire car and trailer to unpack. You might as well forget about looking glamorous, at least for now. Besides, it's not like we're back in D.C. No C-Span cameras here!" he is referring to her attempt to fix her hair after the long trip. This suggests that he thought Libby was somewhat vain and too concerned about her appearance. The other answer choices are not supported by the passage.

8. **The best answer is G.** Prior to the lines mentioned in the question, the passage states that Libby "knew it [the future] could be anything we wanted it to be." The passage goes on to state that Libby, "felt a real thrill of excitement" after she looked around the apartment. These statements best support answer choice G.

9. **The best answer is C.** The first two sentences of the first paragraph set the tone for the paragraph; as far as Libby, the main character, was concerned they had made it "just in time." Additionally, after she got out of the car, her mood began to improve, which suggests that being in the car had made her somewhat unhappy. Answer choices A and D are too general and broad, while answer choice B is too specific.

10. **The best answer is J.** The primary focus of the passage is the young couple's move from Washington, D.C. to Ohio, where their lives would be quite different. They packed up the life they had in D.C. and moved it halfway across the country to the Midwest. The statement "unload our life" refers to this significant change.

### PASSAGE II

11. **The best answer is A.** According to the passage, there is a "specific measure, the 'poverty line,'" that can be used to determine poverty. Because the passage refers to this line as being a specific measure, answer choice A is best. The other answer choices are not supported by the passage.

12. **The best answer is H.** As stated in the passage, "The question of poverty is extremely complex." Therefore, one can infer that measuring poverty is a difficult task. The other answer choices are not supported by the passage.

13. **The best answer is D.** According to the passage, the average American family in the 1950s "spent approximately one-third of its household income on food." When one subtracts 1/3 from the whole, 2/3 remains. Therefore, the average American family spent 2/3 of their income on non-food items in the 1950s.

14. **The best answer is F.** As stated in the passage, "While families today spend about 12 percent of their income on food—nowhere near the 33 percent assumed in the 1950s—the cost of important budget items, such as housing, transportation, and health care, has increased dramatically. Orshansky's poverty measure, which only takes into account the ability of a household to provide itself with food, is missing several essential components to be accurate in modern society. With over $60 billion in federal aid tied each year to this guideline, not to mention an additional $260 billion in Medicaid spending, the fact is many Americans are falling deeper into poverty and failing to receive the aid they so desperately need and deserve." This information suggests that Americans are much poorer than they were in the past, which best supports answer choice F.

15. **The best answer is C.** The passage states that, "Orshansky's poverty measure, which only takes into account the ability of a household to provide itself with food, is missing several essential components to be accurate in modern society. With over $60 billion in federal aid tied each year to this guideline, not to mention an additional $260 billion in Medicaid spending, the fact is many Americans are falling deeper into poverty and failing to receive the aid they so desperately need and deserve." These statements best support answer choice C.

16. **The best answer is G.** As indicated in the passage, "Orshansky's statistical measurement was one small part of the federal government's plan to attack the difficult national economic conditions that were hurting millions of Americans in the early 1960s. President Lyndon Johnson labeled the plan the government's "War on Poverty," and it led to such national programs as Head Start, VISTA, and the Jobs Corps." These statements best support answer choice G.

17. **The best answer is B.** The first paragraph of the passage states that, "Poverty is an enduring problem that must be addressed by all modern societies." It also states that, "projects have to be evaluated to see if they are effective. No one can measure the benefits of aid without defining what poverty is, and when someone has been lifted out of it." Based on this information, one can infer that the paragraph aims to detail the importance of measuring poverty in relation to aid disbursement.

18. **The best answer is H.** The passage states the following information: "While families today spend about 12 percent of their income on food—nowhere near the 33 percent assumed in the 1950s …" This statement best supports answer choice H.

19. **The best answer is D.** According to the passage, "Any change in the measured poverty level of a society is an indicator of economic health within that society, and no president has been willing to increase the perceived amount of poverty for a statistical recalculation, no matter how justified. Indeed, some economists say that updating the poverty measure would increase the number of those considered poor, and therefore eligible for government aid, by as much as 2 percentage points." Based on this information, answer choices A, B, and C are all reasons why no president is willing to make a poverty level change.

20. **The best answer is F.** As stated in the passage, "Orshansky's statistical measurement was one small part of the federal government's plan to attack the difficult national economic conditions that were hurting millions of Americans in the early 1960s." Specifically, Orshansky worked to formulate a way to define poverty so that the government could help those who met this definition. This information best supports answer choice F.

### PASSAGE III

21. **The best answer is A.** This passage is written by someone reflecting on her past. Specifically, the author reflects on both her childhood and her college years. According to this information, answer choices B and C can be eliminated. The author mentions Sr. Benedicta, but the author is not Sr. Benedicta. Therefore, answer choice D can be eliminated, making answer choice A best.

22. **The best answer is F.** Initially, the author discusses her childhood yearning to meet J.R.R. Tolkien. Paragraph 3 details this longing. However, after studying at Oxford as a young adult and experiencing Tolkien's life second-hand, the author realizes that she does not have to meet him to enjoy his writing.

23. **The best answer is D.** The author begins the passage detailing one of J.R.R. Tolkien's greatest works. "Of course, today, most people think of him as the author of *The Lord of the Rings*—a monumental work that became an epic film." This statement indicates the author's acknowledgement of Tolkien's talent. Later in the passage, when speaking of Tolkien's personality, the author says, "Apparently, most people have this impression of Tolkien's as a gentle, grandfatherly sort of man, but, unless you were his grandchild, that wasn't actually the case. In person, he was frequently severe and not terribly friendly." This statement describes Tolkien's intimidating personality. Together, this information best supports answer choice D.

24. **The best answer is G.** As stated by the author, "I loved Oxford." Throughout the seventh paragraph of the passage, the author lists several reasons to support this love for the university. The other answer choices are not supported by the tone or details of the passage.

25. **The best answer is D.** In the fourth paragraph of the passage, the author says, "But, growing up in the Midwest, the possibility of traveling to England seemed very remote. Then I discovered that Tolkien had died years before I'd even started *The Hobbit*. I forgot about my dream and got down to the business of school and sports and college applications." Based on this information, one can infer that the author came to an understanding of her situation.

26. **The best answer is H.** As stated by the author, "Sr. Benedicta was a very smart but very kindly, elderly nun. She was no slouch as an academic and had published several highly-regarded books in her field. As a colleague, she had spent time with Tolkien when she was newly hired at St. Stephen's College. One day, when she had asked me how I liked studying at Oxford, I decided to tell her about my dream." The author's decision to share this dream was an effort to connect to Tolkien through Benedicta, which best supports answer choice H.

27. **The best answer is B.** The author begins the passage with the following, "I was introduced to Tolkien's writings by a friend when I was 10 years old. Aileen gave me a copy of *The Hobbit*, and told me her father was reading *The Lord of the Rings* to her and her brother at the dinner table every night after the family had finished eating. I read *The Hobbit* and was hooked. By the time I was 14, I had read every piece of fiction Tolkien had published." Based on this information, one can infer that the author very much enjoyed Tolkien's writings and she continued reading his works. This best supports answer choice B.

28. **The best answer is F.** The author says that she loved "the stories our housekeeper would tell of Lawrence of Arabia's ghost who, apparently, lived in our own quarters. When inexplicable drafts would sweep my room, our housekeeper swore it was Lawrence." One can infer from this statement that the housekeeper thought Lawrence of Arabia's ghost haunted their quarters.

29. **The best answer is C.** In the passage, the author says that Sr. Benedicta "was no slouch as an academic," followed by the statement that the professor had "published several highly-regarded books in her field." Based on this information, one can infer that Sr. Benedicta was an excellent researcher.

30. **The best answer is J.** The passage states, "Oxford isn't set up like most American universities. It's not a single uniform entity. Instead, it's a collection of 39 independent colleges, each with its own internal structure and activities, with an overlying administration that conducts examinations and confers degrees." As it is used in this statement, the word *uniform* means consistent.

### PASSAGE IV

31. **The best answer is A.** By reading the passage, one learns about El Niño and its effects. For example, the first paragraph ends with the following statement: "Of these, probably the best-known is the El Niño-Southern Oscillation, popularly called 'El Niño.'" Following this, the article offers details about the oscillation, such as, "Along this area of South America, El Niños reduce the upwelling of cold, nutrient-rich water that sustains large fish populations. Predators such as larger fish and sea birds depend on these populations for survival, as do local fisheries." The passage also details the various political ramifications of El Niño. This information best supports answer choice A.

32. **The best answer is G.** According to the passage, "The term El Niño was first reported in scientific circles in 1892." Because 1892 is within the 1890s, answer choice G is best.

33. **The best answer is C.** The paragraph starts with the following: "Sometimes, pressure caused by intense weather can have unexpected political effects." The paragraph then goes on to

give examples of how the weather may have caused certain political events, such as the French Revolution and Germany's failure to capture Moscow during WWII. This information best supports answer choice C.

34. **The best answer is F.** Paragraph 2 of the passage puts the following information in parentheses, "The reverse phenomenon, a cold ocean current, is known by a corresponding term, La Niña, Spanish for 'little girl.'" While this information is interesting and related to the topic, it is not immediately relevant to its surrounding context.

35. **The best answer is D.** The paragraph states, "Sometimes, however, the various oscillations "beat" together at the same frequency, causing the fluctuations to be synchronized. When this happens, scientists say the resulting weather can be intensified." These details best support answer choice D.

36. **The best answer is H.** The third paragraph of the passage begins with the following: "As climatology developed as a discipline, scientists discovered that both trends in the current were part of a larger phenomenon affecting global climate patterns, the Southern Oscillation. The definition of El Niño has therefore expanded and continues to change as climate researchers compile more data." The paragraph then goes on to detail the process by which this definition is determined, which best supports answer choice H.

37. **The best answer is A.** When detailing El Niño, the passage states, "Most of the time, strong El Niños bring wet winters to the Southwestern United States and milder winters to the Midwest. They tend to bring dry conditions to Indonesia and northern Australia." Of the four given answers, answer choice A is most closely related to these weather effects. Because California is in the Southwestern United States, unusually wet weather indicates an El Niño weather pattern. Furthermore, because droughts are reported in Australia, it seems likely that an El Niño oscillation is present. The other answer choices are not supported by the passage.

38. **The best answer is G.** According to the passage, "Some scientists argue that unusually cold weather brought by a strong El Niño phenomenon caused significant crop damage in 1788–89, which many say contributed to the French Revolution. Other climate researchers claim that strong oscillation coupling, combined with strong El Niños in the late 1930s and early 1940s, led to a profound cold snap in Northern Europe in the middle of the Second World War. The scientists argue that this unexpected cold snap significantly contributed to the failure of Germany to capture Moscow, which changed the course of World War II." This information taken in its entirety best supports answer choice G. Answer choices F and H are not supported by the passage, and, although the passage indicates that some scientists argue that El Niño changed the course of the Second World War, nothing in the passage suggests that Germany would have won had it not been for El Niño.

39. **The best answer is D.** According to the passage, "Weather effects can be damaging. The warming patterns of El Niño are one of the leading causes of natural damage to coral reefs, while wider ENSO fluctuations may cause flooding or drought to occur on land. In these cases, extreme shifts can cause economic pressure by disrupting entire fishing industries or damaging crops." Based on this information, damage to coral reefs, flooding, drought, and the disruption of fishing industries and crops are all possible effects of El Niño. Only answer choice D is NOT mentioned specifically in the passage as a negative consequence.

40. **The best answer is J.** The last paragraph of the passage states that, "ENSO phenomena, along with the other three oscillations, are separate from those attributed to global warming. The causes are completely independent." The passage also indicates that "because El Niño and global warming both can result in strong temperature variability, disruptive rain distribution, and extreme damage to a variety of ecosystems, any synchronicity will be closely observed by scientists seeking to document the total effects of each." Based on this information, once can conclude that, while they are independent of one another, El Niño patterns and global warming have similar effects of global weather, making answer choice J best.

## Science Reasoning Test Explanations

### PASSAGE I

1. **The correct answer is D.** According to Scientist 1, "Recent studies by researchers at Northeastern University and the University of Victoria may suggest that dark matter … does not actually exist." Additionally, Scientist 1 goes on to say that, "a galaxy, seen collectively, has its own gravity and essentially drives its own rotation at a constant rate." This information best supports answer choice D.

2. **The correct answer is G.** The process of elimination is a good approach to correctly answering this question. Scientist 2 believes that dark matter is responsible for the warped shape of galaxies such as the Milky Way, so eliminate answer choice F. Both scientists conclude that not all questions about the universe have been answered, which supports answer choice G. Only Scientist 2 believes that cosmic microwave radiation suggests the presence of dark matter, and neither scientist believes that dark matter can be easily observed in the universe, so eliminate answer choices H and J.

3. **The correct answer is B.** According to the passage, Scientist 2 believes that dark matter can explain many cosmological phenomena. Therefore, you can eliminate answer choice A. Scientist 2 does not indicate that dark matter is visible using modern instruments, only that its effect on other objects in the universe can be detected; eliminate answer choice C. Scientist 2 does not suggest that dark matter was ever a part of the general theory of relativity, so eliminate answer choice D.

4. **The correct answer is J.** Both scientists mention the existence of warped galaxies in their discussions regarding the existence of dark matter in the universe. Clearly, this phenomenon is critical to their viewpoints, so eliminate answer choices F and G. Scientist 2 references Newtonian physics in order to suggest the existence of dark matter in order to explain the additional gravitational pull needed to account for warped galaxies, so eliminate answer choice H.

5. **The correct answer is B.** According to Scientist 2, "Without the existence of dark matter, the Magellanic Clouds would not have sufficient mass to have such a strong effect on the bend of the Milky Way galaxy." If it were discovered that the Magellanic Clouds are actually more massive than previously thought, Scientist 2's theory would be weakened. The other answer choices are not supported by the passage.

6. **The correct answer is F.** The main point of difference between the two positions is whether dark matter exists. Scientist 1 believes that it does not, while Scientist 2 believes that it does. The other answer choices are not supported by the passage.

7. **The correct answer is A.** According to Scientist 1, the new research suggesting that dark matter does not actually exist is based upon Einstein's theory of general relativity. If Einstein's theory were flawed, Scientist 1's position would be seriously weakened. The other answer choices are not supported by the passage.

### PASSAGE II

8. **The correct answer is F.** According to the passage, "Contact fungicides, because they do not penetrate the plant tissue, are more easily washed off the leaf by rain. This results in a shorter residual control period and more frequent re-application of the fungicide." Table 1 indicates that *boscalid* is a contact fungicide, so it should be applied more frequently than the others listed.

9. **The correct answer is C.** The passage states that, "Fungicide applications should not be initiated after the R5 growth stage." There is nothing in the passage to indicate that *chlorthalonil* should not be applied during any growth stage.

10. **The correct answer is G.** According to the passage, "research has shown that the most critical time to protect soybean plants with fungicides is from the R1 through R5 growth stages. Fungicide applications should not be initiated after the R5 growth stage." This information best supports answer choice G.

11. **The correct answer is D.** According to the passage, "ASR infections generally begin in the lower leaf canopy … Coverage as dense as 400 spray droplets per square inch is considered ideal." Therefore, you can predict that if fewer than 400 droplets of any fungicide were applied, the chances of the plant becoming infected with ASR would increase.

12. **The correct answer is H.** According to Table 2, systemic fungicides are most effective because they are absorbed into the leaf tissue and translocated throughout the plant. Table 1 indicates that *myclobutanil* is a systemic fungicide. Because the question asks you to put the fungicides in order

of increasing effectiveness, *myclobutanil* should be last on the list. Eliminate answer choices F and J. The data indicate that contact fungicides are the least effective and that *boscalid* is a contact fungicide. Therefore, *boscalid* should be first on the list.

13. **The correct answer is B.** According to the passage, "research has shown that the most critical time to protect soybean plants with fungicides is from the R1 through R5 growth stages. Fungicide applications should not be initiated after the R5 growth stage." Because growth stage R6 is characterized by a full seed, fungicide should NOT be applied during the full seed stage.

## PASSAGE III

14. **The correct answer is H.** According to the passage, "sea cucumbers are wormlike and tend to burrow." The echinoderm pictured most resembles a worm so it is likely a sea cucumber.

15. **The correct answer is C.** According to Table 1, sea lilies are *crinoids*. Table 2 shows that sea lilies are most likely found in offshore mud and ooze, which best supports answer choice C.

16. **The correct answer is F.** According to Table 1, the feeding method of *halothurian* (sea cucumbers) is to swallow mud and sand; therefore, it is most likely that sea cucumbers burrow into the mud and sand in search of food. The other answer choices are not supported by the passage.

17. **The correct answer is B.** According to Table 1, *asteroids* (starfish) are predatory and use their tube feet to move. Table 2 indicates that starfish can be found on the deep-sea floor. This information best supports answer choice B.

18. **The correct answer is G.** According to the passage, "sea cucumbers are wormlike and tend to burrow." Table 1 indicates that sea cucumbers feed by swallowing mud and sand, digesting any organic material, and ejecting whatever is left over. Sea cucumbers in large numbers could, simply by feeding, alter the physical and chemical composition of the sea floor. The other answer choices are not supported by the data.

## PASSAGE IV

19. **The correct answer is C.** The passage states that the average starting weight of the guinea pigs in each group was 50 g. According to Table 2, the guinea pigs in Group 5 had the highest average weight after 8 weeks (98 g). Guinea pigs in this group were given Feed V.

20. **The correct answer is J.** According to Table 1, the guinea pigs that were fed a grain-based feed with vitamin supplements (Group 2, Feed Q) gained an average of 32 grams over 8 weeks ($82 - 50 = 32$). Therefore, on average, these guinea pigs gained 4 grams per week ($32 \div 8 = 4$).

21. **The correct answer is C.** According to the question, the guinea pigs in Group 9 would receive Feed V (the same as Group 5) and a vitamin supplement. According to Table 2 and the information in the passage, the guinea pigs in Group 5 had the greatest average weight and height after 8 weeks. Additionally, the guinea pigs given a vitamin supplement (like in Group 6) showed more growth than the guinea pigs that did not receive vitamin supplements (such as Groups 7 and 8). Therefore, if Group 9 is given Feed V with vitamin supplements, the guinea pigs will most likely be even larger than those in Group 5, which best supports answer choice C.

22. **The correct answer is J.** The best way to answer this question is to examine each of the answer choices, and eliminate those that are not supported by the data in Table 2.

    **Answer choice F:** Group 7 is the control group. Group 6 was given Feed W. It is not true that the guinea pigs in Group 6 are twice as long as the guinea pigs in Group 7 so eliminate answer choice F.

    **Answer choice G:** Group 5 was given Feed V. It is not true that the guinea pigs in Group 5 weigh three times more than the guinea pigs in Group 7, so eliminate answer choice G.

    **Answer choice H:** Group 8 was given Feed Y; guinea pigs in this group did not have the greatest average length, so eliminate answer choice H.

    Answer choice J is correct, because it is true that the guinea pigs in Group 7, who received Feed X, are similar in both weight and length to the guinea pigs in Group 8, who received Feed Y.

23. **The correct answer is B.** The only difference between Group 3 (the control group) and Group 4 is that the guinea pigs in Group 4 were fed fruits and vegetables in addition to a grain-based feed. The students can hypothesize that those fruits and vegetables must have a relatively low nutritional value, since the guinea pigs in Group 4 did not grow any more than did the guinea pigs in the control group.

24. **The correct answer is F.** According to the results of the experiments, the guinea pigs in Group 8 gained the least amount of weight ($74 - 50 = 24$ grams) and had the smallest increase in average length ($23.25 - 20 = 3.25$ cm).

## PASSAGE V

25. **The correct answer is B.** To answer this question, find the symbol for 403°K in the key, then find 2 hours on the $x$-axis. Move your finger up from the 2-hour mark until it intercepts the appropriate symbol. You will see that the symbol is at approximately 50% on the $y$-axis.

26. **The correct answer is J.** To answer this question, find the symbol for 423°K in the key, and then follow its movement on the graph. You will see that conversion stops at 8 hours, so eliminate answer choice F. The line is steepest from 0 to 2 hours, which suggests that this is the period during which most of the conversion took place.

27. **The correct answer is A.** In each experiment, 5% Ru/C catalyst was used. The temperature was varied in each experiment, and the conversion rates were dependent on the various temperatures. Two different types of acid were used.

28. **The correct answer is J.** The data in Table 1 and Table 2 indicate a direct relationship between time and percentage of conversion. As time increased, at each temperature the percent of acid converted also increased up to a certain point, at which time conversion stopped.

29. **The correct answer is A.** The conversion of propionic acid was studied in Experiment 2, the results of which are shown in Table 2. According to Table 2, the conversion of propionic acid at 403°K stops at 8 hours, which does not support the chemist's claim.

30. **The correct answer is H.** The results of Experiment 2 are shown in Table 2. To answer this question, find the symbol for 423°K in the key, and then follow its movement on the graph. Notice that at 12 hours, the PA is approximately 80% converted. It makes sense that, if the conversion continued, 100% of the acid would be converted sometime between 12 and 14 hours.

## PASSAGE VI

31. **The correct answer is D.** The passage states that, "Levels near properly adjusted gas stoves are often 5.0 to 15.0 ppm and those near poorly adjusted stoves may be 30.0 ppm or higher." Because Home 1 has CO levels greater than 25.0 ppm, it is the most likely home to have a poorly adjusted gas stove.

32. **The correct answer is G.** According to Table 1, Home 2 has CO levels between 15.0 and 25.0 ppm. The passage states that average CO levels in homes without gas stoves are between 0.5 and 5.0 ppm, which means the CO levels in Home 2 are above average.

33. **The correct answer is A.** The passage states that, "Levels near properly adjusted gas stoves are often 5.0 to 15.0 ppm and those near poorly adjusted stoves may be 30.0 ppm or higher. CO levels between 0.5 and 15.0 ppm are considered safe." This information best supports answer choice A. The remaining answer choices indicate safe levels.

34. **The correct answer is H.** According to Table 1, Home 4 has CO levels between 1.0 and 5.0 ppm. The passage indicates that levels between 0.5 and 15.0 ppm are considered safe. Therefore, the CO levels in Home 4 are not dangerous, so if it has a gas stove, the stove is properly adjusted and does not have to be removed.

35. **The correct answer is C.** According to the passage, "CO levels between 0.5 and 15.0 ppm are considered safe." Therefore, a CO level of 10.0, as found in Home 6, will not pose any danger to the residents.

## PASSAGE VII

36. **The correct answer is G.** According to the passage, Plot B received applications of high-nitrogen fertilizer, whereas Plot A received applications of low-nitrogen fertilizer. Both Table 1 and Figure 1 indicate that the leachate from Plot B always contained higher average nitrogen levels than the leachate from Plot A. This information best supports answer choice G.

37. **The correct answer is A.** The passage states that, "The leaching of high concentrations of nitrogen into natural waterways can throw off the environmental equilibrium of the aquatic ecosystem, often resulting in an increase in plant growth that can have a negative impact on the native fish populations." Therefore, if low-nitrogen fertilizer is applied, any leachate into surrounding waterways should have low concentrations of nitrogen, and the native fish populations will not be affected.

38. **The correct answer is H.** According to Table 1, the average nitrogen concentration in the leachate from both plots during 2003 was higher than in previous or subsequent years. Therefore, more nitrogen was leached from the soil in 2003, suggesting that this year experienced heavy rainfall during the months of the study.

39. **The correct answer is B.** According to Table 1, only Plot A had average nitrogen concentrations less than 10 mg/L. Because this plot received a fertilizer application containing 98 kilograms of nitrogen per acre, you can conclude that the Environmental Protection Agency would consider this application safe.

40. **The correct answer is H.** According to the passage, the nitrogen levels in the fertilizer applied to Plot B were reduced in 2004. However, Table 1 shows that the nitrogen concentration in the leachate from Plot B was 11.8 mg/L, which is still considered unsafe. In 2005, the concentration had gone down to a safe 8.2 mg/L. Therefore, you can conclude that it will take more than one year to reach safe nitrogen levels in the leachate. The other answer choices are not supported by the data.

## Writing Test Explanation

Because grading the essay is subjective, we've chosen not to include any "graded" essays here. Your best bet is to have someone you trust, such as your personal tutor, read your essays and give you an honest critique. If you plan on grading your own essays, review the grading criteria and be as honest as possible regarding the structure, development, organization, technique, and appropriateness of your writing. Focus on your weak areas and continue to practice in order to improve your writing skills.